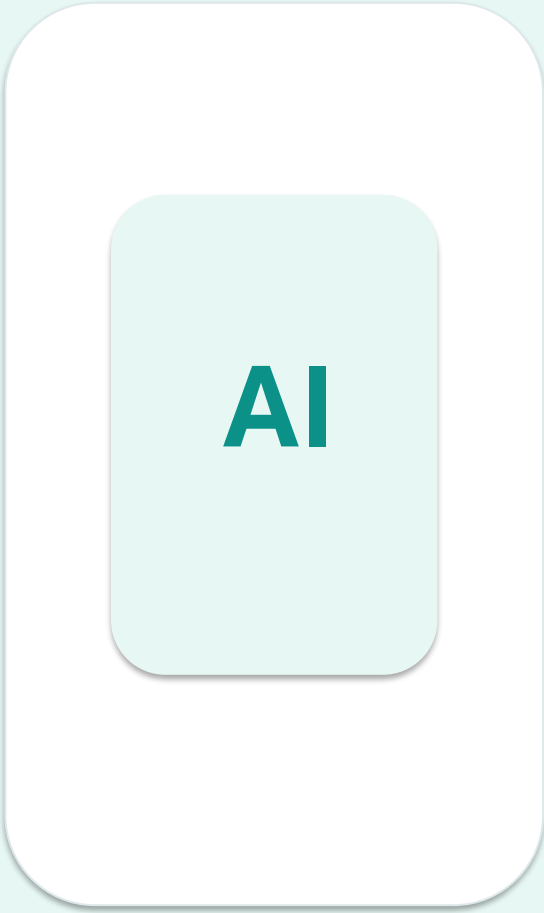
IASYN AI

Intelligent rehabilitation. Measurable progress.

AI-assisted physiotherapy, live movement analysis and clinician-connected recovery monitoring



FIRST PROJECT REVIEW | OCTOBER 2026

THE QUESTION

What happens after a patient leaves physiotherapy?

“Who checks whether rehabilitation exercises are being performed correctly at home?”

The gap: a prescribed exercise is not the same as a monitored exercise.

PROBLEM + USERS

The problem is a gap in continuous rehabilitation

Patients at home

Limited immediate feedback on form, range of motion, repetitions and consistency.

Physiotherapists

Cannot continuously observe every home session or easily review comparable movement metrics.

Caregivers & families

May struggle to support exercise routines without clear progress or tracking information.

Why it is serious: poor visibility into home exercise performance can delay recognition of difficulties and weaken follow-up.

IMPACT

Why this matters beyond the clinic

Impact on users

- Uncertainty about exercise performance
- Difficulty noticing progress over time
- Less actionable information between visits

Potential social impact

- More accessible home-based monitoring
- Better continuity between visits
- Support for remote and resource-limited settings

Our aim is to support clinicians and patients—not replace professional assessment.

NOVELTY + INNOVATION

What makes IASYN AI different?

Live movement intelligence

Pretrained body-pose detection combined with exercise-specific joint-angle, repetition and movement-timing analysis.

Connected care loop

One workflow linking patient exercises, longitudinal metrics and clinician review.

Multimodal roadmap

Explainable AI, interactive 3D pose visualization and optional ESP32 LCD/LED feedback.

Innovation is in the integrated rehabilitation workflow—not a claim that pose estimation itself is new.

SOLUTION

Our solution: IASYN AI

01

Clinician-approved plan

→

02

Guided camera session

→

03

Live movement metrics

→

04

Saved progress + review

Starting use case: seated knee extension with live knee-angle estimates, repetition counting and session history.

TECH STACK

Technology designed for a practical prototype

Frontend

React 18 · TypeScript · Vite

Tailwind CSS · Recharts

Backend

Python · FastAPI

SQLAlchemy · SQLite

Computer vision

MediaPipe Pose Landmarker

Landmarks · angles · reps

ML datasets

UI-PRMD (real motion)

RehabMotion (synthetic, optional)

3D + explainability

Three.js / React Three Fiber

Evidence-grounded AI (planned)

Hardware

ESP32 · USB serial

16×2 LCD · LEDs (planned)

SYSTEM WORKFLOW

From camera input to meaningful review



MediaPipe estimates body landmarks; our algorithms derive motion metrics. Dataset-trained exercise recognition is a later, separately evaluated component.

REVIEW READINESS

Feasibility and progress so far

Completed — Phase 1

- ✓ React/TypeScript application foundation
- ✓ Light-themed healthcare UI
- ✓ Patient, doctor and admin portal shells
- ✓ 24 routes and reusable components
- ✓ Reported successful TypeScript/build checks
- ✓ GitHub push reported: 89dc883

Next — working prototype

- → Phase 2: secure FastAPI login + database
- → Phase 3: real webcam rehabilitation engine
- → Later: dataset-backed ML evaluation
- → Later: clinician reports, AI and ESP32

Feasible because we reuse pretrained pose detection, start with one exercise and validate each component phase by phase.